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Architecture for Regenerative Urban Life: Co-creative Design in Stockholm Royal Seaport, Sweden

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Abstract

This paper accounts for the empirical application of a co-creative methodology based on regenerative design; a theory-based process anchored in ecological principles from landscape architecture and systems ecology. In urban contexts, regenerative design can help recreate and enhance urban ecosystems' ability to generate useful, nature-based, and sustainable products and services. It can also provide high social and aesthetic values. The aim of this paper is to explore how co-creation may assist in operationalizing regenerative design empirically in a formal planning context. For this purpose, a case study was conducted in Stockholm Royal Seaport, Sweden, a modern urban district with official sustainability ambitions. Based on regenerative design criteria, this entailed a solution-oriented design process in collaboration with the City of Stockholm, emphasizing how material structures (tectonics) could be utilized to enable biological and ecological processes (tropism). A design studio constituted the empirical basis for the work, in which facilitated discussions and sketching was performed by a group of planning professionals, supported by a professional illustrator. Emerging ideas were translated into sketches of potential interventions that would establish regenerative processes at the physical site. The result is a comprehensive proposal to address current challenges and future possibilities, visions towards architecture for regenerative urban life.

1. Introduction

Sustainability goals are particularly central to cities, as urban settlements require a range of resources; through local and global circulation of energy, materials, food, water, and waste (cf. Bartek et al. 2025; Bergquist et al. 2020; Hough 2004; Maassen et al. 2020; Russo et al. 2014; Zhifeng et al. 2014). This implies a need for planning and site design that minimizes ecological impacts of urban life both locally and globally, while also providing conditions for a good life, high social and aesthetic values. As argued by Murphy (2005), however, the design of most contemporary cities has instead resulted in both unhealthy ecosystems and citizens. One possible explanation to this shortcoming, is the conceptual separation of humanity from nature; a common feature in urban planning, that brings with it substantial difficulties in recognizing, and hence effectively planning for, our settlements´ dependence and impacts on the environment. Increasing our understanding of the relationship between nature and society is thus fundamental for designing truly sustainable cities.

In the field of urban sustainability and design, diverse knowledge on the human-nature divide, as well as interrelatedness co-exist; from utilitarian perspectives to more relational, more than human, and more justice and egalitarian approaches (Gudynas 2014; Hedfors and Florgård 2012; Langemeyer and Connolly 2020; O'Conner and Kenter 2019; Whatmore 2006). Local and indigenous perspectives to some extent challenge western views; combining geographic considerations with social and symbolic ideas and beliefs. For example, some indigenous communities call nature as the "mother" (Mother nature), implying a balanced and respectful relation with it (Kimmerer 2013; Melo 2013). In urban design, such socio-spatially based approaches can contribute to a more inclusive and pluralistic understanding of nature, including respect for nature's cycles and dynamics (van den Berg 1990). In this paper, we make

use of a formal planning context in Stockholm, Sweden, to explore the potential of co-creation and regenerative design to contribute planning perspectives that are more aligned with site-specific conditions as well as the notion of designing with Mother nature.

1.1. Aim and objectives

The aim of this paper is to explore how co-creation may assist in operationalizing regenerative design empirically in a formal planning context. The objective of this case study, therefore, was to synthesize design solutions enabled by architectural structures and local ecological processes, with human needs at the neighborhood level. Based on a set of criteria for regenerative design developed by Bergquist and Hedfors (2018), this approach has earlier been used theoretically in Uppsala, Sweden (Bergquist et al. 2019). However, the application described in this paper, is the first time the methodology is tested embedded in a formal planning context. In contrast to the detailed design of individual objects, which is common for example in architecture, the focus of regenerative design as applied in this study is mainly on relationships between people, the built environment and landscape functions (nature); visible as well as non-visible connections that are relevant from a broad (global) systems perspective and have the potential to regenerate resources locally. Accordingly, sustainability is defined as regenerative functions and processes at multiple scales and in tangible as well as intangible and relational dimensions.

1.2. Design with Mother Nature

By switching prepositions between the words Design and Nature, completely different relationships between human activities and their surroundings are easily generated (Hedfors and Florgård 2012). The prepositions *for*, *as*, *after*, *of* generate perspectives that in different ways imply a distancing from nature. For example, Design-*for*-Nature implies that we think we can represent nature, and at the same time place ourselves outside. *As* is classified as mimicry, *after* means we follow nature and repair it, while *of* becomes a dominant approach. The origin of such reasoning is Design-with-Nature (McHarg 1969) – starting from a design based on a deep understanding of how nature works – an engaged and integrative relation with nature closely connected to the idea of Mother nature. Hence, explicitly aiming to design *with* nature, may conceptually guide urban designers to go beyond an anthropocentric perspective, instead engaging more actively with collaboration between species and symbiosis with nature (cf. Margulis 1971).

Similarly, and as a response to contemporary challenges such as climate change and sustainability, two interrelated concepts have been gaining interest: Nature-based Solutions (NBS) and Nature-based Thinking (NBT). NBS is an umbrella term to describe different interventions *using* nature to resolve societal problems, such as urban forestry, sustainable urban drainage systems (SUDS), deculverting and additions to blue-green infrastructure networks (Barona et al. 2020; D'Arcy 1998; Konijnendijk et al. 2005; Kozak et al. 2020; Wild et al. 2024). NBS can bring benefits to society in multiple forms, such as reduced flood risk, water pollution, air pollution and heat island effects. However, because of its emphasis on *solving* or finding *solutions* to societal problems, NBS has been criticized for being embedded in and reproducing instrumentalized views of nature; reinforcing the idea that nature is external to human

beings, and that therefore it can be exploited (Eggermont et al. 2015; Schröter et al. 2014). Meanwhile, NBT addresses this problematic notion, by engaging with a more systematic and egalitarian relation with nature, based on the understanding that nature and humanity are indissolubly connected (Mercado et al. 2023; Randrup et al. 2020). Consequently, NBT is similar to designing with nature (Hedfors and Florgård 2012; McHarg 1969); though phrased as nature *with* people – not *for* people (Mercado et al. 2024).

In designing synergetic relationships between nature and humanity in the built environment, the concepts of tropism and tectonics (Hedfors 2014) offer additional insights. Originating in landscape architecture theory, tropism and tectonics are defined as two fundamental principles of space formation. Tropism is the plant's striving and movement for light, water, nutrition, and anchoring, which leads to dynamic space-forming growth according to species-specific patterns. Meanwhile, tectonics is instead based on static pillars; supporting elements that hold up carried elements, beams, into a fixed space-forming framework (Hedfors 2014). A crucial difference is the use of living material versus non-living. These two principles can interact in different ways. For example, tectonic structures can support tropism, i.e. regenerative processes such as plant growth, fruiting and harvesting by being climate and weather equalizers. A sunny façade can offer conditions for trellised fruit orchards, and in a greenhouse all conditions can be manipulated.

While architecture that favors regenerative processes may be thought of as regenerative architecture, this notion does not refer to the tectonic properties of architecture, which in themselves cannot be regenerative. Instead, it implies utilizing non-living material structures (tectonics) to enable biological and ecological processes (tropism) that would not otherwise be possible. In striving towards making use of such opportunities, regenerative design, as conceived by Bergquist and Hedfors (2018), is a theory-based design methodology anchored in ecological principles from landscape architecture by Murphy (e.g. 2005; 2016) and systems ecology by Odum (e.g. 1994; 2007). It departs from and reinforces site-specific and positive interactions between humans and the environment.

Specifically applied in urban built environments, regenerative design can be used in densification and new exploitation projects to identify opportunities to recreate and enhance the ability of urban ecosystems to generate useful, nature-based and sustainable products and services, as well as high social and aesthetic values. Design solutions are prioritized that favor human development, while preserving and enhancing, and making use of, ecological values. To design with nature, in this paper regenerative design is combined with other related concepts such as NBS/NBT, tropism and tectonics, representing the first occasion this theory-based design methodology is tested empirically in a formal planning context.

2. Methods

The concepts drawn upon in this paper raise the critical question of how to connect theoretical ideas and generate successful results in planning and design practice. To move from the conceptual to the practical, creative design methodologies are pivotal to effectively bridge the gap between theory and

design (cf. Bergquist et al. 2019). Many successful social sustainability outcomes have been associated with innovative approaches such as co-creation (e.g. Hernández-García 2013; Puerari et al. 2018). In general, co-creation, an established approach in urban participatory planning (e.g. Condon 2008; de Laval 2014), refers to processes to explore, design, implement and evaluate solutions collaboratively (Melles 2021), fostering social learning where different stakeholders enrich a base of contextual knowledge (Hernández-Garcia and Pico 2023). Based primarily on principles of inclusion and participation, it can contribute to practical results that reflect the real needs and desires of communities (Moser and Korstjens 2022). It is about integrating different knowledges, different knowledge cultures, including perceptions and values, and combine them creatively to respond to complex societal challenges (Hernández-Garcia and Pico 2023). In addition, co-creation seeks to foster innovation, allowing creative solutions to emerge from collaboration between multiple actors (Leask et al. 2019). Consequently, co-creation also forms new relationships between participants, the places, and other stakeholders (Soini et al. 2023). As demonstrated by Kiss et al. (2023), co-creation may thereby strengthen and diversify social sustainability outcomes, such as knowledge mobilization, social learning, enhanced sense of belonging and greater motivation for environmental stewardship.

In this paper, co-creation was applied to enable combination of regenerative design theory, tacit knowledge, adaptation to site conditions, and planner practitioners' concrete ideas. It was operationalized through a design studio, gathering multiple participants and engaging collaboratively. Emerging ideas and visions were translated into sketches of potential interventions that would establish regenerative processes at the physical site, thus assisting in planning towards the establishment of architecture that favors regenerative processes, i.e. regenerative architecture. Furthermore, this work included transforming the resulting visions into a comprehensive design proposal, comprising practical illustrations for further use in the official planning process. As opposed to identifying goal conflicts and competition around land use, this approach emphasized potential synergies for enabling regeneration, by use of nature-based solutions and designing for beneficial connections between different elements (specifically tropism and tectonics), landscape features and people in the built environment, i.e. designing with nature. This represents a holistic approach to current and future possibilities, and generated sketches with simultaneous anchoring in theory, local needs, possibilities and conditions, as well as feasibility with regards to technical, legal, and financial limitations.

2.1. Case study

As Stockholm develops, there is a need to transform former industrial properties – brown fields – into new residential areas for the city's growing population. Stockholm Royal Seaport (Norra Djurgårdsstaden) is one such area and was appointed by the Stockholm City Council to be a forerunner in sustainable urban development (Stockholms stad 2025a). In recent years, the city of Stockholm has been in an expansive development phase within this case study area, of which Valparaiso is the neighborhood subject to regenerative design as outlined in this paper (59°21'00.6"N 18°06'25.8"E). A dominant feature of the area is Värtahamnen harbor and close contact with the Baltic Sea. Valparaiso is

in an intensive expansion phase, and there is a desire for new housing, but also service facilities, offices, and park environments, as well as a rail-bound transport link through the area.

2.2. Design studio and co-creation

A design studio based on site-specific conditions and the official planning context in Norra Djurgårdsstaden formed the empirical basis for the work. It was organized as an activity partly connected to the project Transformative capacity in energy, food, and water (TANGO-W), an applied research project that uses the concept of Urban Transformative Capacities (UTC) to evaluate cities' potential for sustainability. Specifically, this implied a particular focus on possible design solutions at the intersection of food, energy, and water systems. The participants in the design studio were therefore selected based on their complementary competences in these fields. Together representing a range of professions from all phases of the planning process, engaging professionals active in formal planning in the city of Stockholm also provided a stronger mandate amongst participants and increased the likelihood of turning resulting ideas into practice. In a series of workshops, this group co-created solutions as detailed illustrative sketches; exploring how local infrastructure, waste flows, ecosystems and other resources in the district could be used as a productive force in the built environment. Potential synergies were identified, through various combinations of energy, water, and waste systems for local food production, as well as other NBS, blue-green infrastructure included. The resulting solutions were then combined to form an overall vision and design proposal for future development.

In practical terms, the design studio was carried out as two workshops of half a day each in collaboration between the City of Stockholm, the Centre for Health and Sustainability (CHS) at Uppsala University, and the architectural firm Tengbom. A project management team representing these organizations prepared and organized the work during the workshops, facilitated the discussions and the visionary illustration work. An illustrating landscape architect documented the process by translating ideas and oral accounts into landscape sketches and illustrations (Fig. 1, left). The number of participants was deliberately kept low with twelve individuals and mandated participation in both physical meetings. Additionally, two invited specialists in aquaponics and green architecture, respectively, were engaged by giving short presentations on the latest developments in their respective field of expertise.

Between the workshops, the project management team met to review the raw material produced, ensuring that all ideas were included in illustrations. Raw sketches were refined, synthesized, and taken to the following workshop for discussion and further elaboration with all participants (Fig. 1, right). When the workshops were completed, the material was finalized, and all sketches were summarized graphically in a comprehensive design proposal.

3. Results

Ideas that arose during the design studio were discussed and thus self-evaluated by participants; in terms of appropriate form and function in relation to site conditions, e.g. regarding the area's history,

ecology, existing and expected architecture and other infrastructure, as well as legislation and finance. This exploratory process was supported by the landscape architect who continued illustrating by turning words into images, with the aim of arriving at consensus for creating a coherent design proposal that visualizes both desirable and realistic interventions, concrete and feasible visions for use in official planning. In this paper, these design possibilities are presented as results. They include sketches from two workshops, illustrations from the processing in between, and results in the form of synthesis sketches which were combined to summarize the proposal for regenerative urban development in the district. Below, each illustration is described in detail with respect to specific design elements as well as their functions and regenerative potential, based on the information verbally expressed by the design studio participants.

3.1. Output from workshop 1

Workshop 1 began with an introduction to the case study area to be co-creatively designed. Discussions then took place and were translated into images as interpreted in real-time by the illustrator. The following illustrations are edited based on the raw sketches produced in this way (see Fig. 1 for an example of how the raw material could look like).

3.1.1. Glass balconies combining private and public spaces

Due to the design studio being tied to the Tango-W project, priority was given to ideas at the intersection of food, energy, and water systems. During discussions, this translated into exploring the potential to produce food locally. As the climate in Stockholm does not permit year-round production, thoughts soon arose of using glass balconies and façades to create habitat for vegetation by trapping waste heat from buildings and incoming sunlight. This way of using the structure of walls and walkways to make room for plants, implies a strong connection by which tectonics enable tropism. Glass balconies are similar to greenhouses, which means that all conditions for plant growth can be regulated all year round. Though the initial aim expressed by participants was solely to enable food production in this way, more greenery inside buildings also implies high aesthetic and recreational values. A challenge following this approach, however, is that the need for maintenance is high, for example for irrigation. This led to the conclusion that rather than letting residents care for plants themselves, some organized supervision is probably necessary for maintaining both production and tidiness. The solution that emerged from the discussions was to combine private and public spaces, i.e. not to functionally separate living areas from the green structures and assigning maintenance responsibility to the property manager. Figure 2 demonstrates how such a setup could be designed.

By placing growing beds closest to the glass façade, plant access to sunlight is maximized and used to drive photosynthesis. At the same time, greenery acts as shading; a NBS that makes use of vegetation when plant growth is the highest during spring and early summer. Consequently, the shading effect increases as does the need for it, offering incremental protection from sunlight and excess heat during the hottest summer months. Conversely, during winter months, the absence of green curtains allows

sunlight to heat the space. It also works as a buffer capturing heat escaping from apartments, thus adding climate equalizing properties to the benefit of both plants and people, in addition to reducing the need for external energy to heat apartments.

The growing areas are made semi-public, which facilitates access for both residents and managers. A form of general loft corridor is therefore created, horizontally connecting individual apartments, which incentivizes residents to walk through the indoor orchards and spontaneously interact with neighbors. To enable both a large inflow of sunlight, and space for taller plants such as smaller trees (e.g. bananas, peaches, citrus), the loft corridor is designed on two floor levels. Apartments are equipped with either indoor balconies or patios, with views and direct access to the public garden/park environment.

Another NBS resulting from this setup is that greenhouse façades reduce material requirements for walls of the residential building, which translates to using less energy and materials for construction as well as maintenance. An unanswered question resulting from this idea is whether this may compensate for the additional energy and material use implied by constructing the greenhouse façade in the first place. Whereas answering such questions would require technical energy-material analysis, this reasoning by the participants illustrated how NBT can help to think beyond the solution and explore other types of interventions, synergies and multifunctionality.

3.1.2. Local waste management

Local management of waste streams may relieve the burden on municipal waste, water, and sewage facilities. A common NBS for this purpose, is to use the bio-remediation capacity of plants for purifying water. Though proven efficient and cost efficient elsewhere (e.g. Romero 2021), arguably this only partially utilizes the potential of introducing plants to local waste management systems. For example, using plants as bio-filters also implies self-provision of essential nutrients as inputs for cultivating food crops and greenery. By adding the use of waste to produce compost, conditions for urban agriculture can be improved (cf. Torres Granados and Calderon Montenegro 2021). Indeed, a common theme during design studio discussions, was to make use of local waste for regeneration of resource inputs to food production, and at the same time, thereby also improving social dynamics. Figure 3 illustrates how this could be achieved in terms of the simultaneous regeneration of clean water, food and spaces for social interaction.

Integrating food and waste systems locally, as envisioned in Fig. 3 also limits negative externalities associated with the need for mechanical and chemical water purification provided by municipal amenities, and food imports. Instead, the building – i.e. the tectonic structure – is here quipped with infrastructure for catching, storing, and circulating storm water, as well as black water. By installing urine-separating WCs and/or vacuum toilets, urine may be redirected for use as fertilizer for crops, while faeces are transported to the basement level for secure composting. Biological waste from rooftop and indoor cultivations is also disposed of and circulated in this way, along with household waste. However, to enable this setup, the tectonics of the roof needs to support additional weight, to provide habitat that develop tropism (e.g. deep enough soil, combined with for example lightweight biochar skeleton soil).

The building also generates a footprint on the ground, which obstruct the natural potential for plant life. To some extent, the roof compensates for the footprint. Calculations of the Biotope Area Factor (BAF) estimates approximately 20% compensation from this setup (Stockholms stad 2025b). The combination with glass façades may compensate more, i.e. by maximizing productive space.

With this setup, the different waste streams are composted and combined in different ways in the property's own soil factory in the basement. Adding paper waste (e.g. cardboard) enables the production of growing substrate with the appropriate carbon/nitrogen balance. The finished compost soil is used to fill up growing beds in rooftop and indoor cultivations. Urine and nutrient-rich liquids from the composting processes (e.g. compost tea) are used to locally produce nutrient solutions for hydroponic systems. Dry, less nutritious materials, such as wood chips from the maintenance of adjacent parks, is used as growing media for mushroom cultivation in basements and other dark environments. Local wastewater treatment and rainwater harvesting implies that less water needs to be supplied by the municipal system, thus combining values such as proper use, reuse, conservation and regeneration.

During the design studio discussions, the possibility of handling additional waste from the cruise ships arriving at the dock daily, was also considered. This would imply that the district contributes valuable ecosystem services to the shipping industry as well. This service may be commercialized, as it partly contributes to climate adaptation of the cruise tourist industry and means that the shipping companies do not have to dispose their waste through conventional municipal waste systems. However, the amounts of waste needed to be handled in this way would far exceed what is possible to circulate locally in the district. On the other hand, this implies an opportunity to producing a significant surplus of compost soils, growing substrate and nutrients, i.e. by-products that can also be commercialized, for example through sales to other growers in the area who do not have their own composting systems, as well as farmers in Stockholm's outskirts and regional agriculture. To enable this scaling, automated systems may be an option, e.g. with AI and software to ensure that the right resource ends up in the right place at the right time. However, such technologies are associated with additional energy consumption. Further studies are therefore necessary to assess this possibility, which was only briefly discussed by the design studio participants.

Most importantly, for the setup to be feasible, maintaining production requires an actor with formal responsibility for running the operations. In Fig. 3 this has been solved by the roof of the building housing a garden center run by a commercial gardener, who in addition to the building's compost systems, runs a commercial roof garden with fruits, berries and vegetables. Whereas this idea exemplifies an approach to more circular urban metabolism (Lucertini and Musco 2020), it also serves to demonstrate the multifunctional possibilities revealed by applying regenerative design co-creatively in multidisciplinary teams.

3.1.3. Balconies prepared for small scale gardening

Also in more conventional private apartments, balconies may be prepared to enable cultivation. Glass balconies can be compared to greenhouses, which means that all conditions for plant growth can be

regulated all year round. However, since this requires maintenance efforts, during discussions it was clear that it is essential that residents themselves can choose whether, and if so, what, they want to grow. Figure 4 therefore shows a solution where balconies are equipped with a glass façade and growing boxes that can easily be opened or covered with lids, to quickly switch between functions.

In the open position, the boxes can be filled with soil (from the basement soil factory) and used for cultivation. When the crops are not active, and for residents who do not want to grow anything, the lids can be closed, and the boxes instead serve as seating areas with a view of the outside. In the closed position, the space in the boxes can also be used for small-scale composting, especially during winter months, as the residents dispose of their household waste, and hence also become self-sufficient with their own compost soil. Tectonics is here important, as it provides the materiality for this to happen, but also the possibility to connect with people's ideas and expectations in an emancipatory way. Balconies for small scale gardening open many multifunctional possibilities, social and environmental. Co-creative regenerative design here played an important role in identifying this potential.

3.1.4. Green corridors are activated for cultivation, transit, recreation and citizen participation

Outdoor spaces are likewise programmed to enable cultivation, but at the same time are kept flexible to be adapted for activities responding to potential citizen participation. In Fig. 5 this is illustrated by Bremenstråket (Bremen route), a continuous and relatively long route that would also be used for mobility on foot and by bicycle. This illustration shows several possibilities to engage with nature, facilitate social dynamics, and improve the local environment. Green corridors, with urban forestry and gardens, including small scale "farms" with animals provide opportunity for people to actively take part in regeneration.

The hard surfaces are considered tectonic foundations because vegetation cannot grow there. However, they can support nearby plantings through stormwater runoff if tilted in the right direction. It is therefore important that the stormwater is not led away with drainage pipes but steered to benefit the vegetation. Another strategy is to make hard surfaces less hard, and percolating, such as using gravel, and adding nearby tropism. The narrow green corridors can be supported by stormwater permeable tectonic curbs that prevent wear and protect trees and plantations. The image in Fig. 5 shows a tectonic foundation (without erected framework) for bike paths and walkways, without tropism. Where the tectonic foundation is percolating instead, the chicken pen and fruit trees increase the relative share of tropism versus tectonics (the Trop/Tect-ratio). Where the tectonic foundation is used for cultivation, it results in a relatively large proportion of tropism. Compared to the illustration of the building with roof top and greenhouse cultivation (Fig. 3), the design of Bremenstråket (Fig. 5) however implies relatively more tectonics, less tropism. This demonstrates that tropism-tectonics interaction can be thought of as a sliding scale, depending on how possible design solutions are combined.

Here, also ideas on SUDS are relevant to consider, an important part of NBS. By means of collecting water using natural materials and low energy consumption, such as appropriately tilted tectonic

foundations, storm water can be used for different regenerative purposes, for example for watering urban gardens, cleaning streets, and so on. SUDS are usually considered at larger scales, such as deculverting urban streams, parks and green corridors, which is the case here. SUDS can also add functions such as buffering for storm surge overflows, soil percolation, and bio-filtration. In this sense, SUDS, as a NBS can favor regenerative processes, as well as planning to face complex risks such as heavy rainfall resulting from climate change (Woods-Ballard et al. 2007).

The solution in Fig. 5 also includes small-scale urban agriculture in the form of fruit trees planted in an enclosure that also serves as a pen for chickens, kept by the gardener nearby. The setup originally resulted from a discussion on how to separate pedestrians and cyclists, but through co-creative regenerative design, additional possibilities were identified and added. In spring, the chickens provide weed control, and in late summer and autumn, the trees contribute fallen fruit not processed for human consumption, providing automated food supplement and the opportunity to forage for chickens, who also find shade under the foliage during hot summer days. This reduces the requirement for maintenance, while creating habitat for the chickens that allows their natural behavior, and thus also contributes to animal welfare values. In this way, this narrow green corridor is activated and gives reason for people to linger and observe. Activation of the site thereby enhances interactions between passing humans and the immediate environment, creating regenerative ecological functions of importance to humans, animals and plants alike.

Apart from creating eventful and ecologically important green links through the area, this form of cultivation lanes can also promote exchanges between residents and those commercially active in local food production. During the design studio discussions, a desire to demonstrate sustainable food production was clear, which here takes the form of agroforestry with fruit trees and opportunities to carry out experiments, for example, perennial grains and other plants that are not disturbed, but rather benefited, by the presence of chickens and their functions such as weed control and fertilization. The presence of fruit trees also creates habitats for pollinating insects.

3.2. Output from workshop 2

After the first workshop, a larger-scale sketch was created to give an overview of the neighborhood (Fig. 6). This was then supplemented with the detail sketches (Fig. 2–5) and notes from workshop 1, resulting in a first synthesizing sketch (Fig. 7). During workshop 2, this material provided the basis for further discussions and additional sketching. Notes were made to specify, connect and complement the elements and processes already identified.

3.2.1. Fish farming and aquarium

In Fig. 7 detailed ideas, elements and processes have begun to be linked, but the connections were still imprecise, and some gaps were identified. This meant further development of the ideas and connecting parts by adding new elements. The possibility of fish farming was a theme that was discussed particularly carefully during workshop 2. In part, there were discussions about supplementing

horticulture systems on rooftops and in balconies with small scale hydroponics and aquaculture, in circular systems that are then called aquaponics. Mainly, however, the discussion came to concern how larger scale commercial fish farming could be housed under buildings, since on the site there are large underground spaces that were previously used for storage. These nowadays unused “caverns” may provide space for commercial fish farming on a rather large scale. Above ground, the possibility of making parts of the aquaculture systems attractive and open to the public could also contribute to experiences in the area. Since Valparaiso is directly connected to the water, and has a history as a port, the idea arose to create a public museum with a thematic focus on water and the sea. Adding AI/IT-supported infotech for self-guided tours resulted in the site being called an AIquarium (Fig. 8).

The initial idea was to use toilet waste as nutrients to grow fish feed in multiple stages. Such a system has long been in use at the Universeum museum in Gothenburg. The possibility of doing something similar in Valparaiso was discussed, however with a clearer focus on humans' relationship to water and seafood, by showing examples of how keeping fish can also contribute food to people, and how waste management in circular blue-green systems can also make fish farms more sustainable. Location of this visitor center is envisioned in direct connection to the wharf, to benefit from visitors arriving via the cruise ships, thereby activating the site and more clearly connecting the tourism industry with everyday life and the regenerative processes in the district.

3.2.2. Towards an integrated whole

The last part of workshop 2 was mainly dedicated to concretizing how parts identified up to that point could be connected into an integrated and well-functioning whole. Realism and feasibility of practical implementation also became a focus during the closing discussions. The result is Fig. 9, which summarizes central parts of the vision, with a focus on its multiple values and functions.

Buildings in the same block are linked by footbridges. Roof gardens offer recreational opportunities and walking paths for residents in an exciting roof landscape. Cultivation takes place partly in open fields outdoors, and in greenhouses on the roof tops and in connected balconies. In the nearby Finland Park and Plektrum Park, space is provided for recreation of a more conventional nature, as well as community gardens and demonstration sites for inspiration and learning. Agroforestry containing fruit trees, fruits, and nuts, connects to the site's history and existing oak environment in the Finland Park. Multi-layered stands including single conifers, together with agroforestry are the most effective ways to favor tropism while limiting negative consequences of heavy precipitation and rapid snowmelt (Florgård and Palm 1980).

The relationship with Mother nature is enhanced and made visible by arranging and displaying the different elements and processes in a circular way; activities such as cultivation produces food and biomass (waste), the first used for human consumption, while the latter as compost for starting the cycle again. Irrigation needs are met by SUDS and rainwater circulated though buildings. Apart from yielding food crops, the activities also provide social and participation opportunities, and help other species to flourish (i.e. insects, birds).

Commercial production is managed by gardeners, who also have offices, and a garden and visitor center housed in one of the residential buildings. This is mainly made of glass with only certain parts divided into floors, which enables a large inflow of sunlight, and a ceiling height that allows the indoor cultivation of even larger trees, such as banana plants and other tropical fruit. This is another example of tectonic structures that favor tropism, by creating the conditions for growth, fruiting and harvesting of tropical species that would otherwise not be possible in the site. By attracting bees and other pollinating insects, efficiency is further increased. NBS interventions, such as the garden and visitor center, may also bring economic benefits (Wild et al. 2024), as the relatively large scale implies surplus can be monetarized. The site also functions as a meeting place with recreational and therapeutic values through its tropical/mediterranean forest character, clean moist air, openness and visibility from the street. Such environments may help people recover from stress, depression and anxiety, as some studies of the benefits of NBS suggest (e.g. Vujcic et al. 2017).

In the garden center, information is given to the public, courses and theme days are arranged for residents, the public and the district's commercial actors. The building next door houses fish farms and the AIquarium, which creates additional experiences for both residents and long-distance visitors. Overall, this creates a vibrant and exciting mix of both private (residential), commercial (restaurants and food production), public spaces and small-scale agricultural landscapes.

For those interested in growing food themselves, allotments are offered on roofs and in greenhouses. Waste and return heat from nearby Stockholm Exergi are used for heating and enables year-round production of food as well as experiences, recreational and social values. Researchers from universities and commercial actors are also invited to benefit from the innovative infrastructure, for research, experimentation, production and sales directly to consumers in the area. The place teems with life all year round and becomes a well-known landmark and symbol of the ambitious sustainability work and high living standards in Norra Djurgårdsstaden.

3.2.3. Final design proposal

Based on the material presented to this point, the sketches and notes from the design studio were completed and compiled into a comprehensive design proposal (Fig. 10). Important to note is that this design and detailed sketches are neither formally decided nor definitive in any way. Rather, they are to be regarded as a series of possibilities, which have been co-created by combining the perspectives of professionals within the city of Stockholm, anchored in regenerative design and hence empirically exemplifying the potential of establishing architecture (tectonics) that favors regenerative processes (tropism) at the site.

The design of the Bremenstråket connects to the Finland Park, creating a green corridor that enables movement of fauna as well as people in an exciting and productive environment. This creates multifunctionality through a combination of regenerative processes, SUDS, biodiversity, recreational opportunities, mobility and high social and aesthetic values. This functional integration also has an educational value and can therefore be complemented with informative walking routes through the area,

and signage that describes connections and regenerative relationships that may be difficult to see and understand at first glance.

The beneficial interaction between tropism and tectonics is clearly demonstrated by the glass façades and greenhouses; creating a variety of opportunities for both residents and commercial actors to produce food all year round, while the spaces can also be used for education, meetings and relaxation. Additional costs for this infrastructure are partially offset by the fact that glass façades work as noise protection, which allows buildings that are otherwise limited to offices to house apartments instead, in line with zoning regulations. Waste and water management, integrated with food production, also generates a surplus of food and nutrient-rich garden soil. Income from surplus that is sold finances wages for gardeners/urban farmers with permanent employment with the municipality or property owners. As a result, a probable economic calculation is that the costs can be covered by the return made possible by the investment in the regenerative architecture, a possibility already noted in earlier NBS projects (e.g. Wild et al. 2024). During the design studio at least, the assessment by the participants suggested great potential for making the design proposal cost-effective and competitive compared to the alternative of developing the district more conventionally. However, further in-depth studies are needed to specify such financial calculations, explore the legal space in detail, and further concretize the specific design solutions presented in this paper.

4. Discussion and conclusions

Planning for regenerative urban development requires strategies that integrate human activities, the built environment (tropism and tectonics) and local ecology to generate synergies. The co-creative methodology for regenerative design applied in this study enabled the identification of concrete solutions that illustrate potential pathways for integrating ecological and human processes while regenerating resources directly on-site. Although locally specific, these solutions are transferable as methodological examples, demonstrating how theory-based, participatory design grounded in local conditions can translate sustainability objectives into actionable interventions.

In conventional planning processes, aspirations for a sustainable future often remain abstract. Incorporating diverse professional and disciplinary perspectives through structured co-creative design can operationalize these aspirations, moving beyond rhetorical goals. The approach presented here also reflects a relational perspective on the human–nature relationship, in which (mother) nature is valued not merely as a resource but as an active partner with intrinsic worth.

In the original conception, tropism and tectonics are two incompatible principles of space formation. Tectonics is non-living, while tropism is the principle of life among plants. In the past, urban planning has rarely been seen as an opportunity to promote vegetation, horticulture and agriculture. What has been demonstrated in this paper, is how humans can create conditions for living things with the help of non-living materials. This perspective informs the principle that tectonic structures should only be justified when they enhance natural processes (tropism) and contribute to ecological regeneration. Sealed

surfaces, for example, should be offset through design features that strengthen vegetation and other regenerative processes. Regenerative design thereby aims to balance consumptive (degrading) and regenerative (nourishing) processes, shifting from a paradigm of minimizing harm to one of actively reinforcing regeneration.

Unlike conservation, which prioritizes non-interference, regenerative architecture employs built structures to enable and amplify biological growth, reducing the resource appropriation associated with human habitation. This principle holds conceptual significance: it offers a framework for aligning urban form and function more closely with ecological equilibrium. However, considering the substantial overshoot of contemporary human society (Bergquist et al. 2020), this is not the same as to say that regenerative architecture is enough for reaching a zero-sum result. Even the best of regenerative designs would not compensate entirely for all resource use associated with human life. For example, a Swedish urban middle-class lifestyle has been estimated to exceed a fair and sustainable global resource share by a factor of approximately 77 (Bergquist et al. 2020). This massive resource use is also associated with significant climate and other environmental impacts (Bartek et al. 2025), contributing to the transgression of six out of nine planetary boundaries (Richardson et al. 2023). Whereas this justifies substantial improvement and the promotion of more resource-efficient lifestyles, it also serves to demonstrate the enormous challenge associated with any attempt to fitting human society within Earth system boundaries (Rockström et al. 2023). As it is within this context that regenerative design exists, a high degree of humility is appropriate.

While regenerative design cannot by itself eliminate overshoot, it represents a responsible and necessary approach for reducing environmental impact and fostering mutually beneficial human–nature interactions. Its value lies both in its practical capacity to deliver localized ecological improvements and in its conceptual role as a framework for operationalizing the long-term aspiration of urban life in closer alignment with Earth system boundaries.

Declarations

Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

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This work was organized as a collaborative effort within the project “Transformative capacity in energy, food, and water” (TANGO-W), an applied research project that uses the concept of Urban Transformative Capacities (UTC) to evaluate cities’ potential for sustainability. As such, it was partly made possible by funding from the European Union´s Horizon 2020 research and innovation programme.

Author Contribution

Daniel Bergquist and Per Hedfors contributed to the original conception of the regenerative design methodology applied in this study. Case study design, material preparation, data collection and co-creative facilitation was performed by Daniel Bergquist. Per Hedfors and Jaime Hernandez-Garcia contributed theoretical and conceptual development and refinement. Analysis was performed by all authors. The first draft of the manuscript was written by Daniel Bergquist, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Figures

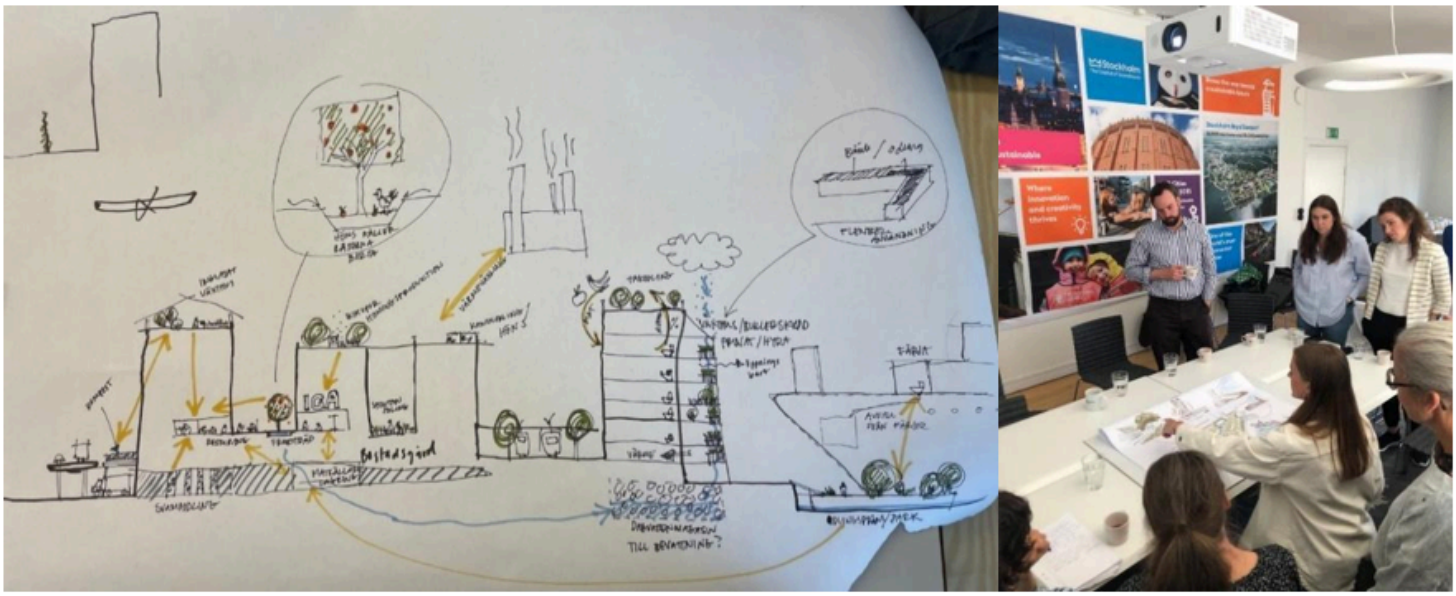


Figure 1

Left: example of sketch under production during one of the design studio workshops. Right: The second workshop at the project office in Norra Djurgårdsstaden. The illustrator shows sketches from the previous workshop to be discussed and improved upon. Photo: Daniel Bergquist

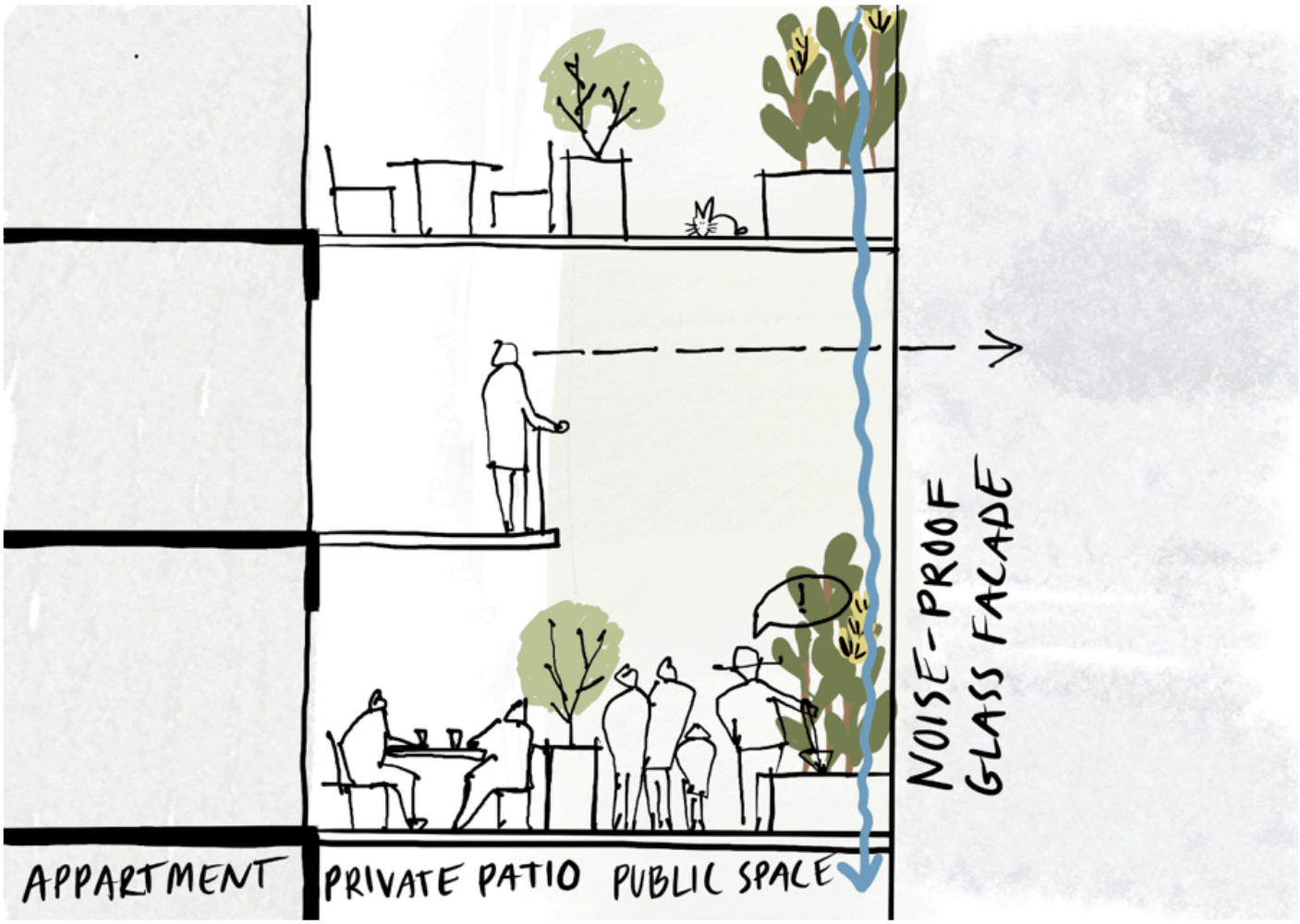


Figure 2

Section showing combination of housing (apartments), with adjacent private and public spaces. Glass balconies/façades provide tectonic structure supportive to tropism, enabling the simultaneous production of food, aesthetical and recreational values in the form of constant greenery and attractive spaces for socializing

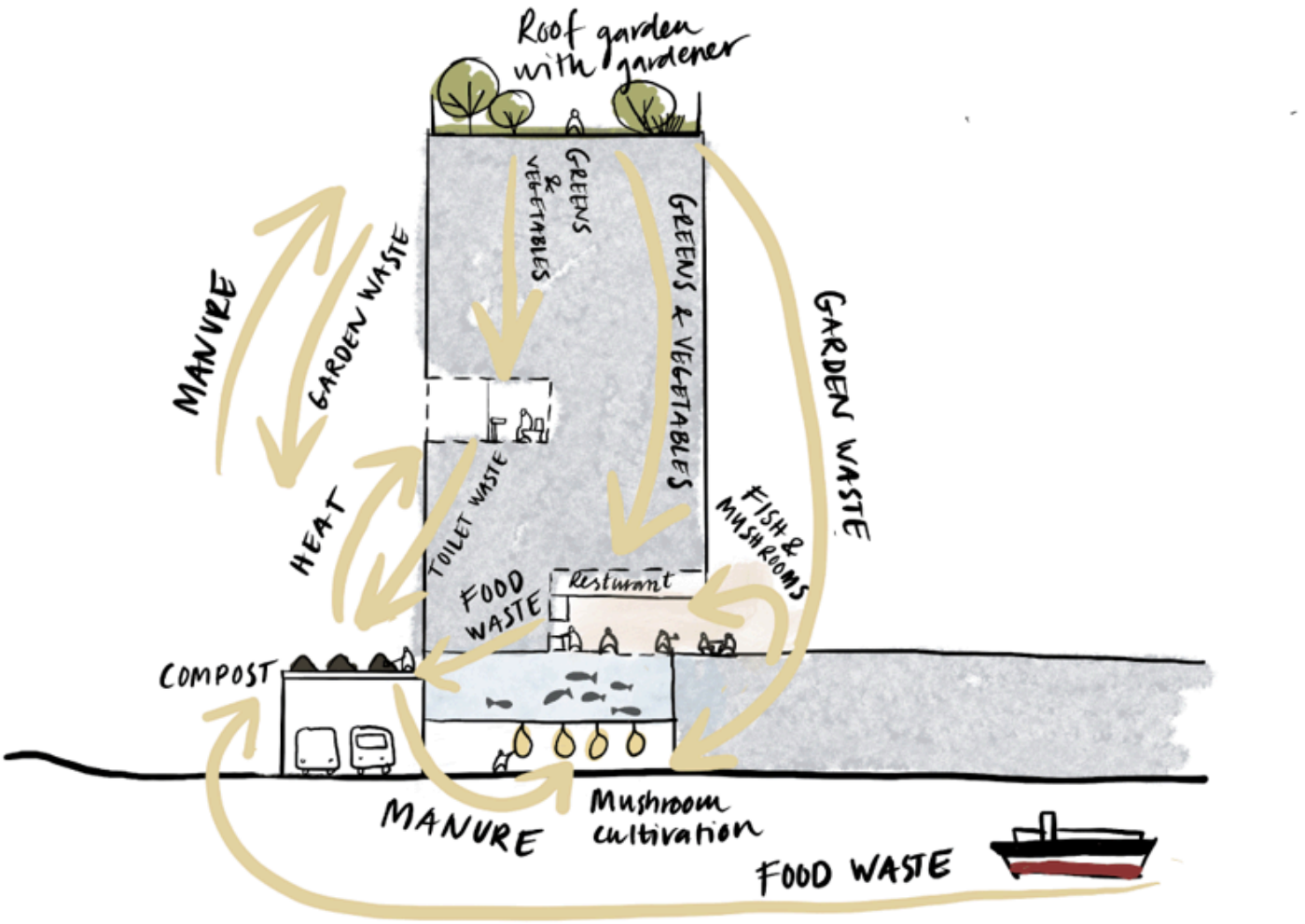


Figure 3

Buildings are equipped with systems for local disposal and circulation of waste streams, for composting and mushroom cultivation in basement levels, and production of growing substrates and nutrients used in rooftop and indoor cultivation



Figure 4

Multifunctional balconies that, through glazing and prepared boxes, enable switching between small-scale cultivation, composting, and seating in an attractive location. Left: winter mode/inactive cultivation, with composting under seating. Right: spring mode, preparing cultivation that will also act as growing green curtain for protection during summer



Figure 5

Vision of Bremenstråket, with a semi-public garden and chicken yard. Functional separation between pedestrians and cyclists, and green corridor for ecosystem services. Activation of residents through voluntary participation in activities organized by the local gardener. Multifunctionality creates interesting views for passers-by and from nearby restaurants

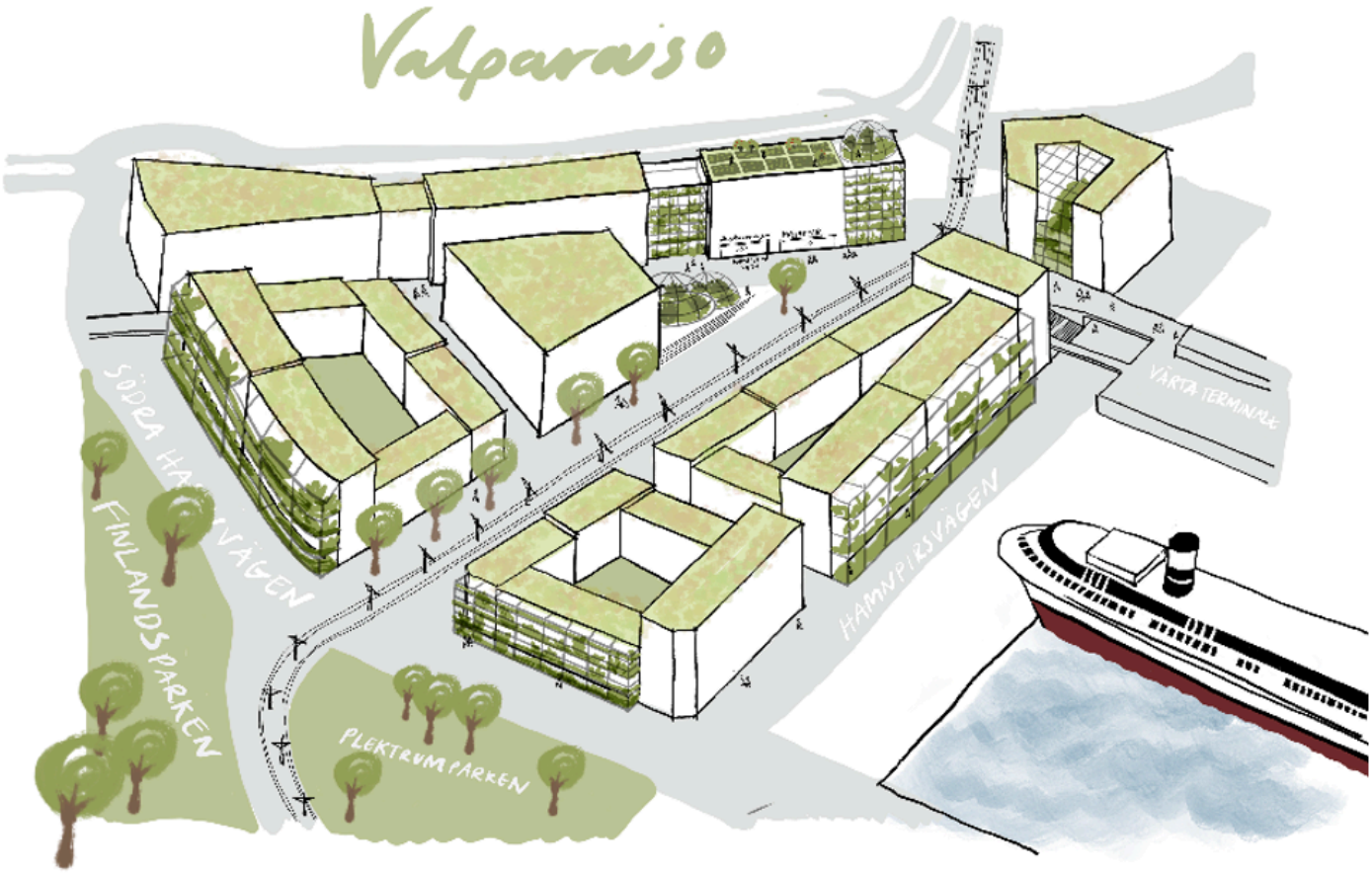


Figure 6

Large scale sketch of the emerging vision of Valparaiso. Glass façades, roof and indoor gardens are shown, yet still without clarifying connections between structures

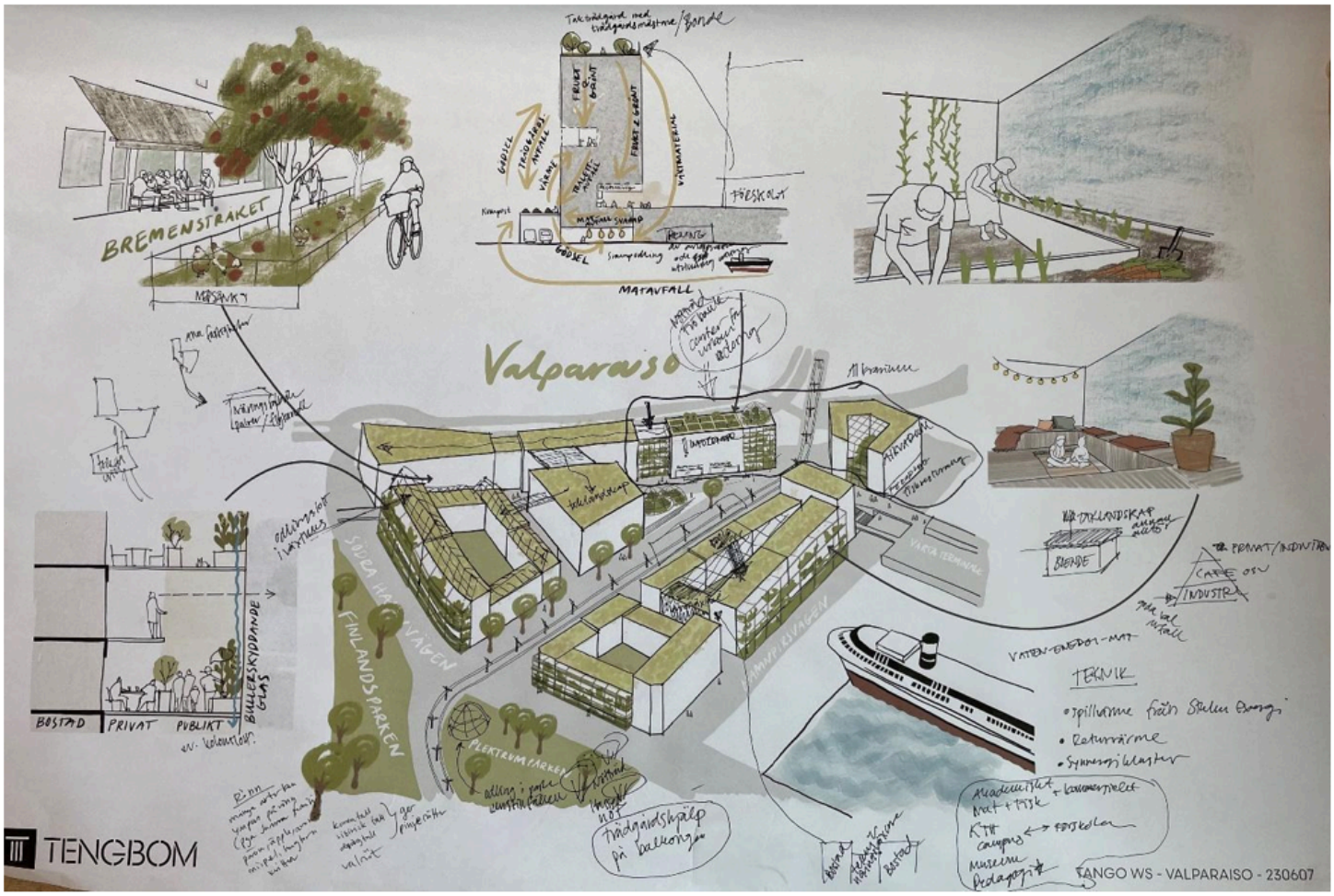


Figure 7

First synthesizing sketch, supplemented with the detailed sketches and handwritten notes from the first workshop. This version was used for presentation to the design studio participants, to initiate further discussions and additional sketching during workshop 2

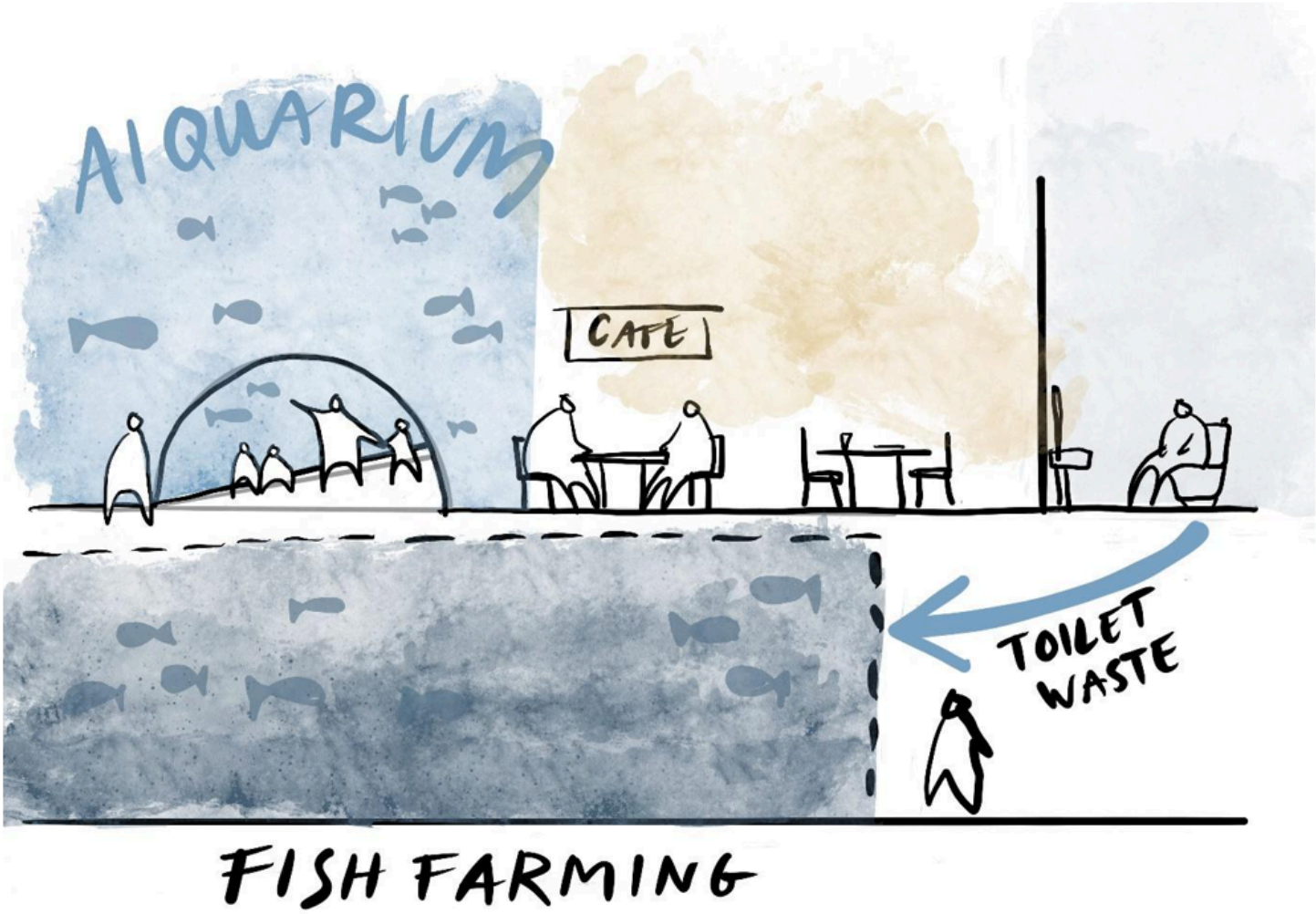


Figure 8

Fish farming and AI-quarium at the quay in Valparaiso connects urban life with tourism, and offers experiences, and activation of the site, as well as opportunities to demonstrate and create insights into the district's experiments with innovative AI/IT-supported infotech, smart VA systems and food focus

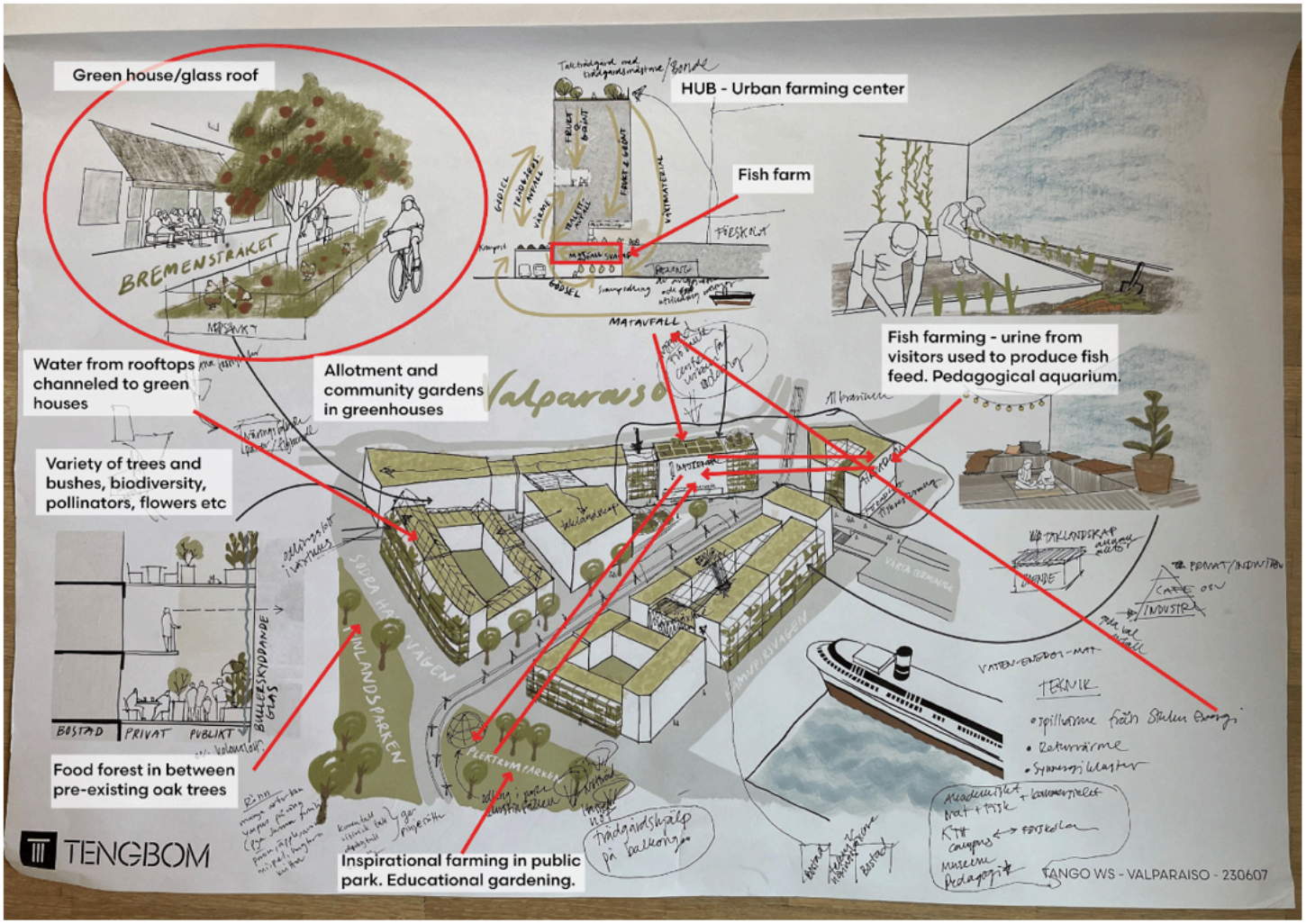


Figure 9

Working sketch with documentation of parts of the discussions during workshop 2. In red, clarifications of the values and functions of the respective subsystems

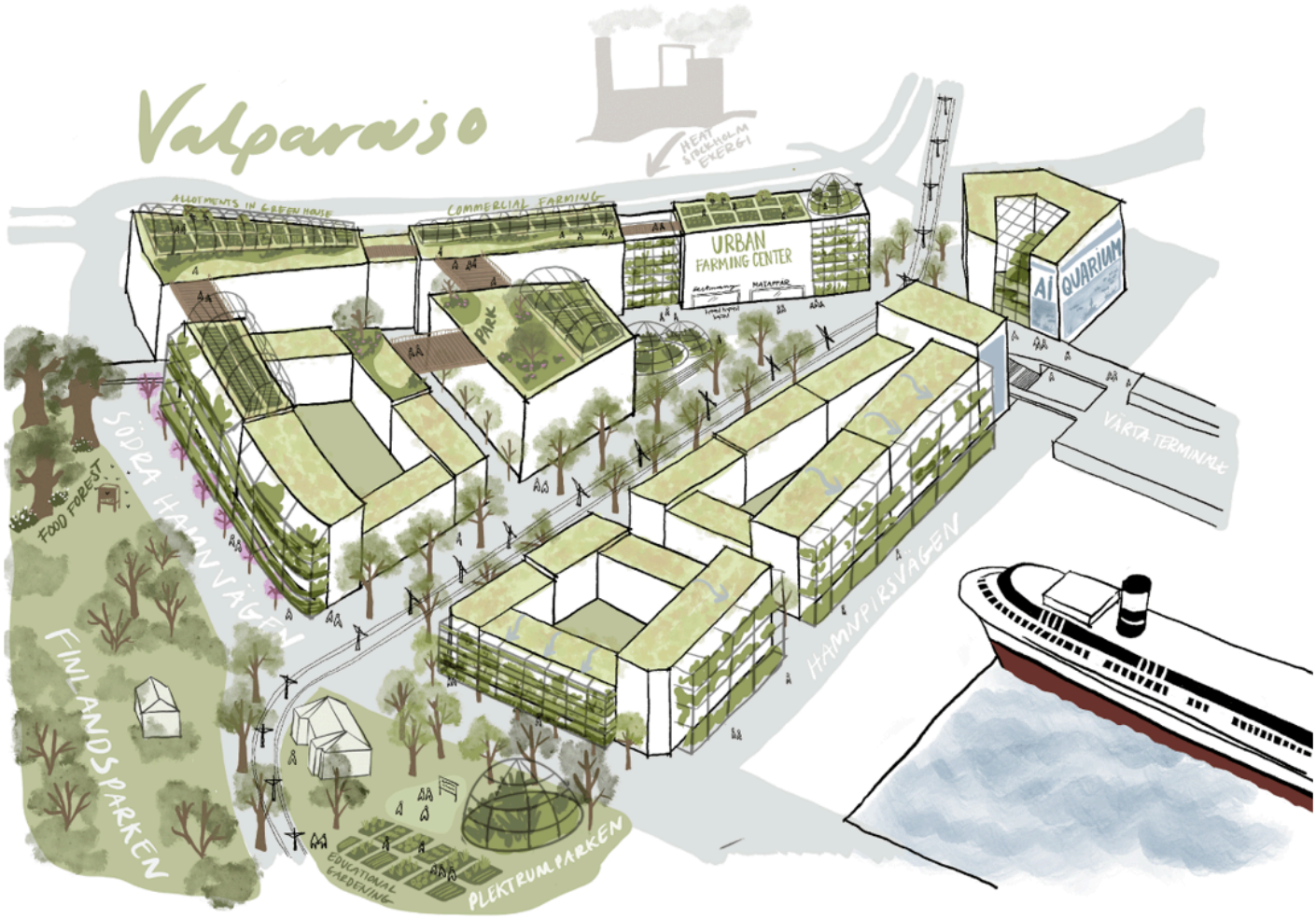


Figure 10

Final design proposal for Valparaiso – possible design solutions to develop the area as a coherent whole integrating human and ecological processes, and regenerating resources directly in the urban fabric